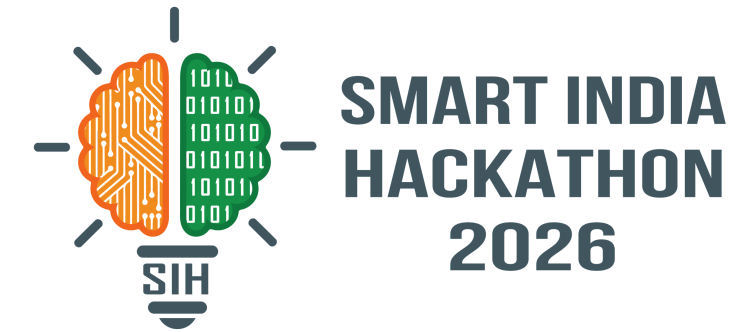
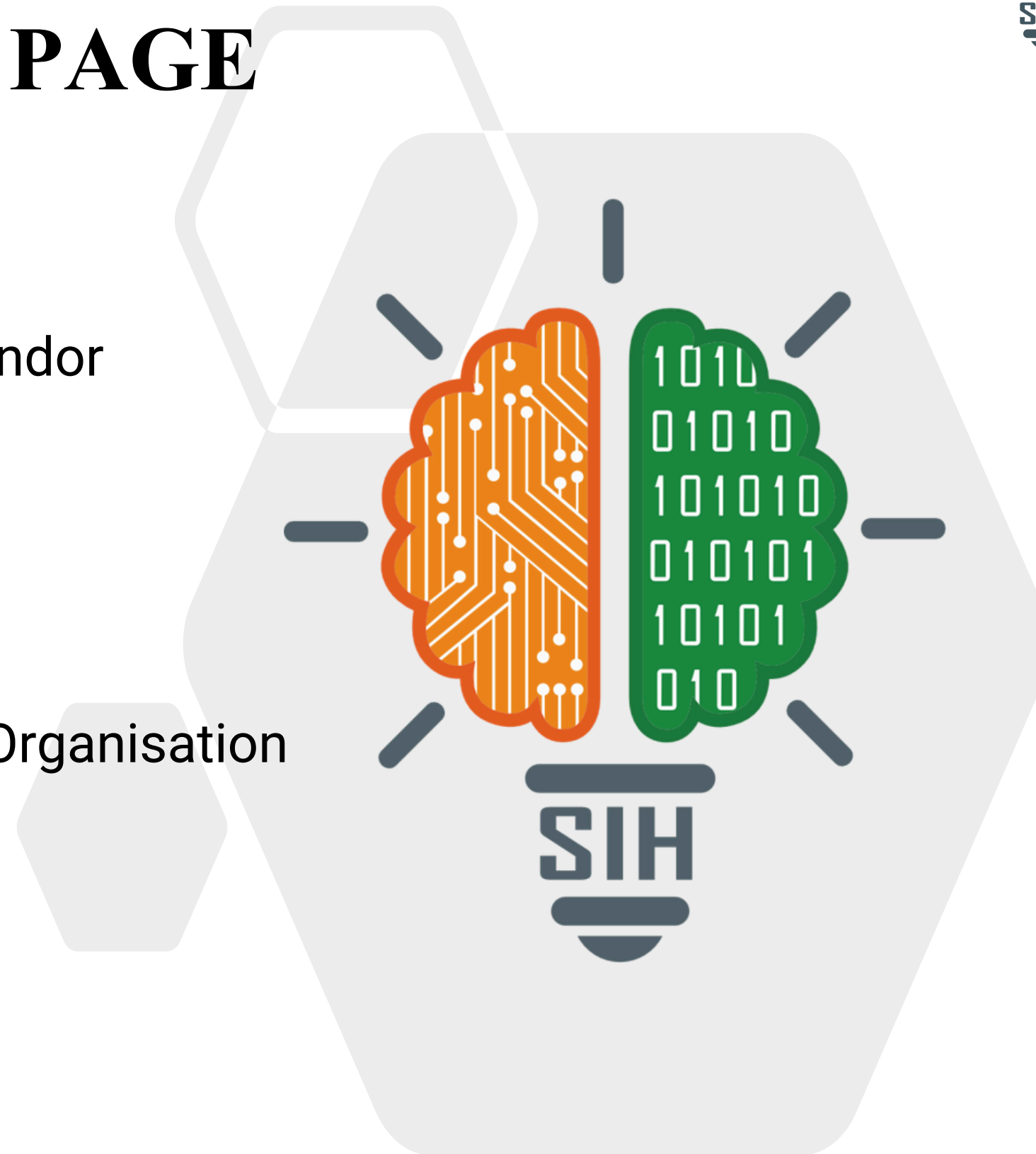


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** - SIH26155
- **Problem Statement Title** - AI-Driven Multi-Vendor
Network Security Compliance Auditor
- **Theme** - Blockchain & Cybersecurity
- **Organization** - National Technical Research Organisation
- **PS Category** - Software
- **Team ID** - 9E5544
- **Team Name** - Team TechTadkaa



Backstory of Problem



Heterogenous Network

Firewalls, routers & switches from **multiple vendors**, each with **different CLI syntax, hierarchies & OS**.



Misconfiguration Risk

Misconfigurations create **insecure** protocols, **weak authentication**, excessive access & poor logging.



Compliance Burden

Must adhere to **CIS, NIST, STIGs & ISO 27001** standard across diverse devices.



Current Pain

- Manual audits - **slow, error-prone**
- vendor tools - fragmented & **locked-in**
- New vendors/OS - **existing** parsers **fail**

46%

of major network-related outages cited configuration/change-management failure

100 +

CIS Benchmarks across 25+ vendor product families

Proposed Solution

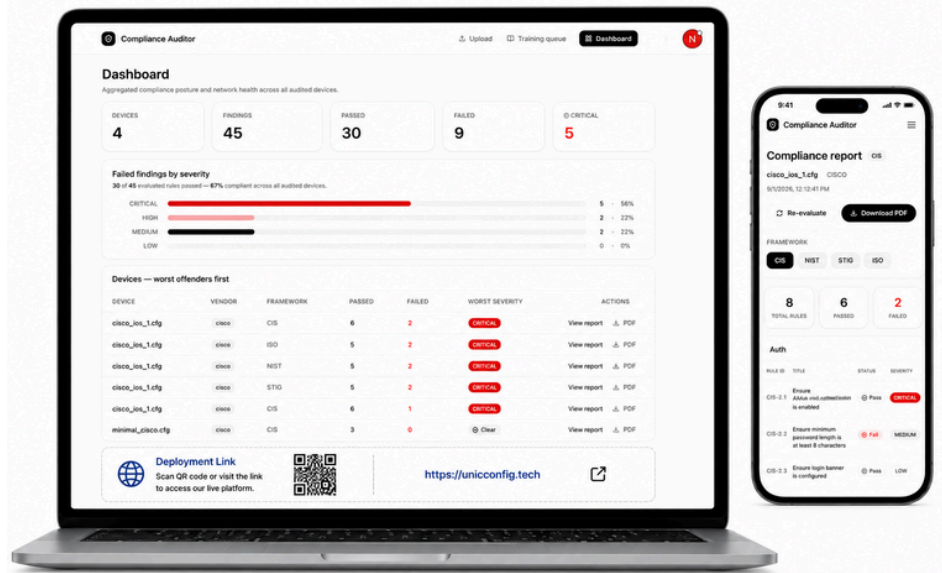


- UniConfig** transforms **heterogenous network** configurations into a **common security model** and compares them against **multiple compliance criteria**
- RAG powered Configuration**
Uses **relevant vendor knowledge** to provide **context aware** configuration data. It also **adapts** to previously unknown configurations
- One Baseline. Multiple Standards.**
It **reduces** fragmented **compliance** checks by evaluating diverse device configurations against **CIS,NIST,STIG** and **ISO 27001**
- Adaptive & Future-Ready**
The system doesn't just support new vendors it **learns** how to support them, **adapting** to **unfamiliar configurations** through **human-guided learning**.



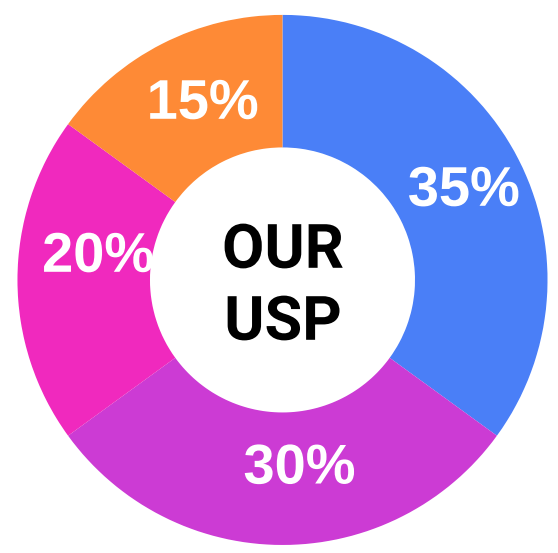
The system doesn't just support new vendors, it can learn how to support them.

Prototype



 **Deployment Link :**

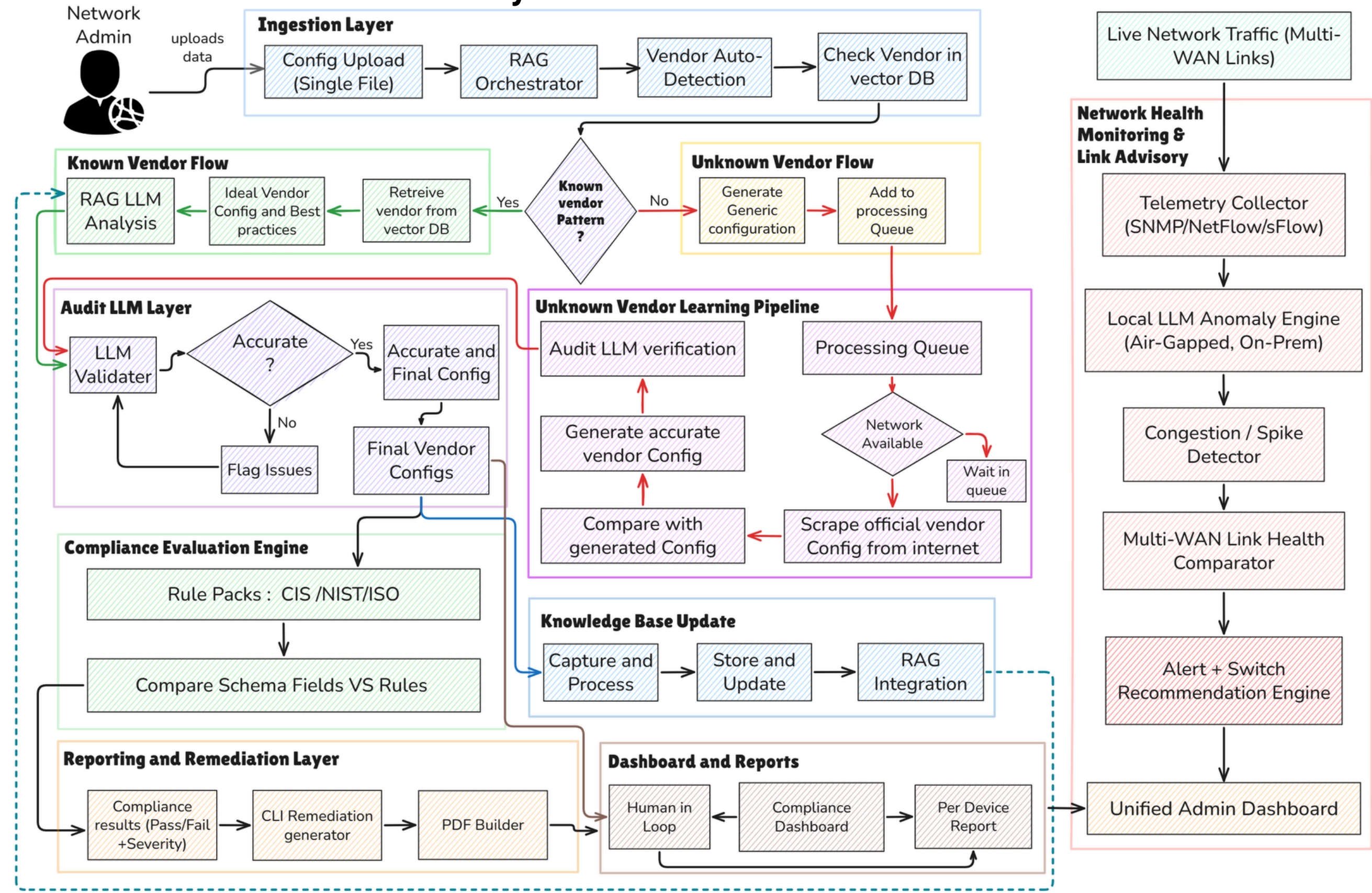
Unique Special Points



OUR USP

-  **Self-Improving Unknown-Vendor Intelligence**
-  **Hallucination-Resistant Configuration Intelligence**
-  **Air-Gapped RAG Configuration Intelligence**
-  **Compliance-to-Remediation Pipeline**

System Architecture



Frontend



Backend



LLM/RAG Layer



Data Layer



Deployment



Network Layer



Feasibility Analysis

Technical:

- It is Built with reliable tools like **LLM/RAG, FastAPI, PostgreSQL and SNMP.**
- **Modular design** makes it **easy to support** new vendors, devices and network traffic.

Operational:

- Automation **reduces manual effort** by handling **configuration checks, compliance evaluation, and reporting.**
- Using **open-source tools and on-premise AI** helps keep the overall **cost manageable.**

Legal:

- Sensitive network data can be processed within the organization, **reducing data privacy risks.**
- The system supports major security standards such as **CIS, NIST, ISO 27001, and DISA STIGs.**

Business Model (Operational Viability)

SaaS/subscription model:

A flexible **subscription model** where pricing is based on the **number of devices and vendors** being monitored.

Add-on revenue:

Custom compliance packs can be provided as **YAML modules** to meet the specific requirements of different industries.

On-prem/air-gapped licensing:

A **one-time licensing option with AMC support** for secure deployments in defense, banking, and government organizations.

Professional Services:

Onboarding and vendor-parser support make deployment easier, even for legacy and specialized hardware.

Challenges and Mitigation Strategies

Multi-vendor complexity

A **hybrid parser** converts different vendor configurations into a common format for **smoother processing.**

AI hallucination configurations

AI-generated configurations are checked by an **LLM validator** and **verified against official vendor configurations.**

Air-gapped environments

A local LLM runs **on-premise**, keeping sensitive network data **secure even in air-gapped environments.**

Scalability

RAG and processing queues help the system easily manage more devices and vendors as the network grows.

Target Audience & Stakeholders



Network Administrators
Manage and configure devices



Security Engineers
Identify and fix security gaps



Compliance/Audit Teams
Verify Security Standards



CISO/Security Leadership
Monitor Security Posture

Potential Impact



Numbers Defining UniConfig



less manual audit effort

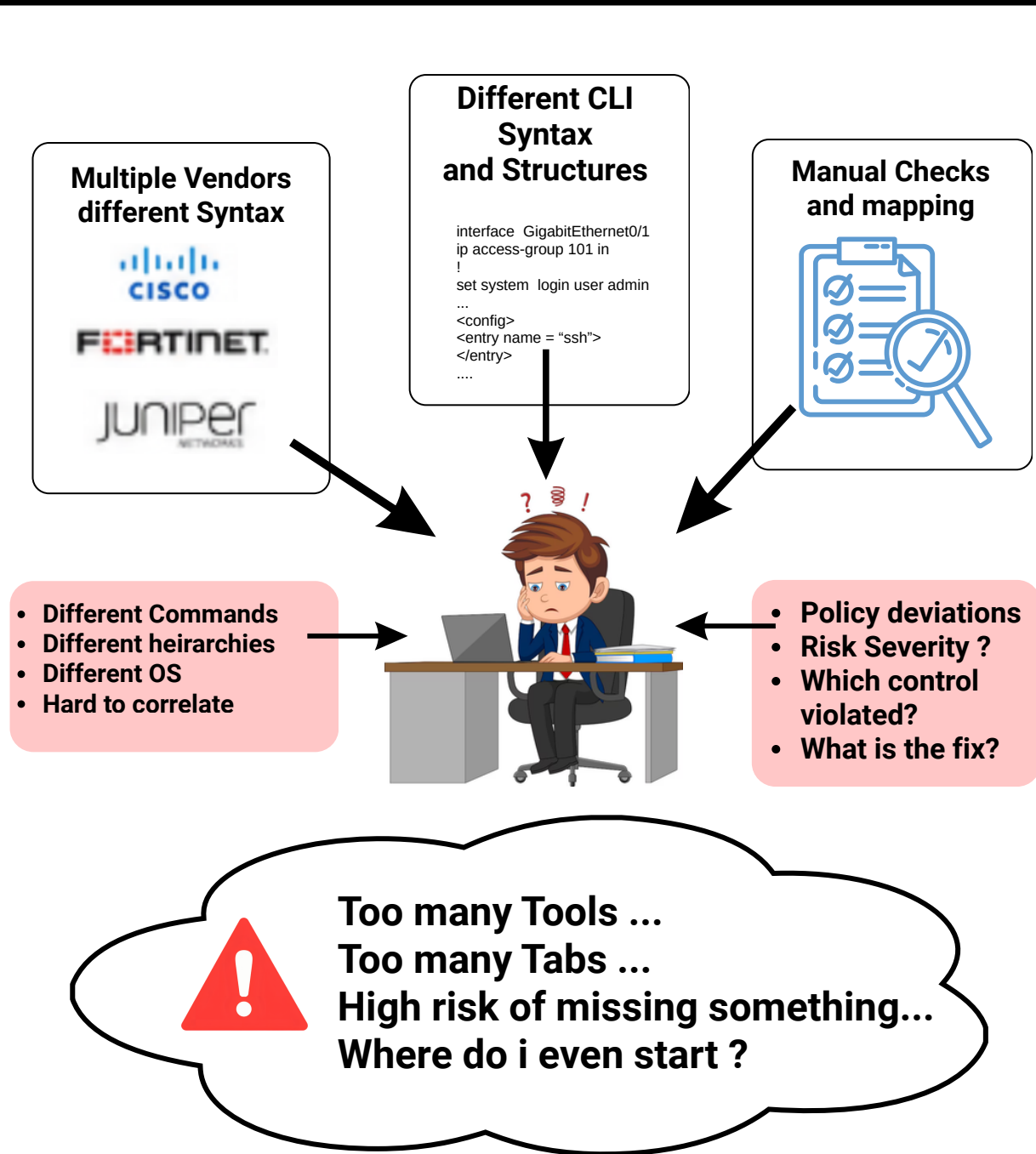


Faster Compliance Preparation



Less Remediation effort

A Scenario showing how UniConfig benefits the target audience



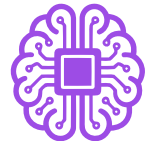
Benefits

How UniConfig Turns complexity into clarity

- UniConfig understands different vendor configurations and converts them into a **common model**
- Creates **per-device compliance reports** with clear **pass/fail** status and **severity** for easier audit review.
- Combines Learnings** of human validation with **reusable** knowledge for **future configurations**.
- Translates **compliance** findings into **clear, actionable fixes** for each device.



Multiple vendors



One Baseline



Multiple Standards

Regulatory and Compliance



1. Aligns with **CIS, NIST, STIG, and ISO/IEC 27001** for cybersecurity compliance.
2. Aligns with the **IT Act, 2000** for protection of systems and digital data.
3. Provides **traceable compliance findings** and reports for **effective security audits**.

Links

[GitHub Repository](#)
[Demo Video](#)



References

- **CIS Benchmarks**
<https://www.cisecurity.org>
- **NIST SP 800-53 Rev. 5**
<https://csrc.nist.gov>
- **DISA STIG Viewer**
<https://public.cyber.mil>
- **ISO/IEC 27001:2022**
<https://www.iso.org>
- **Network Management Market Report 2026**
<https://www.globalgrowthinsights.com>
- **Network Config & Change Management Market Report 2026**
<https://www.researchandmarkets.com>

Research and Identified Gaps

- **Vendor Fragmentation**
Different vendors use different formats, making **cross-vendor analysis** difficult.
- **Fragmented Compliance**
Security controls vary by framework, making **unified compliance assessment** complex.
- **Static & Limited Configuration Analysis**
Existing tools rely on **predefined rules**, making unknown configurations difficult to analyze.
- **Limited End-to-End Remediation**
Most approaches detect violations, but **risk prioritization, remediation** is fragmented.

How we address those Gaps



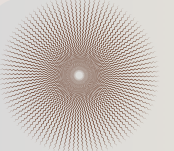
Unified Multi-Vendor Configuration

Normalizes **diverse vendor configurations** into one standard format.



Unified Multi-Standard Compliance

Evaluates configurations against **multiple security standards** in one workflow.



AI Hallucination Mitigation

RAG validation grounds outputs in vendor data and verifies them before final configs.



Intelligent Risk-to-Remediation

Prioritizes risks and provides **actionable remediation** guidance.